

Chatbot with Retrieval-Augmented Generation (RAG)

Apartment Layout Dataset Analysis

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1 Introduction

This project presents the design and implementation of a chatbot system that integrates locally hosted Large Language Models (LLMs) with structured data processing. The system adopts a Retrieval-Augmented Generation (RAG) approach, enabling natural language queries to be translated into structured operations executed on a dataset.

The dataset used in this project is an apartment layout dataset obtained from Zenodo:

<https://zenodo.org/records/7070952>

This dataset contains both geometric and semantic information, including room types, spatial features, floor identifiers, and relationships between apartment entities. The goal of the system is to enable intuitive querying and accurate data retrieval through natural language interaction.

2 System Architecture

The system is implemented in Python, utilizing **Pandas** for data processing and **Ollama** for running LLMs locally. Two models are employed:

- Llama 3.2
- DeepSeek R1

The overall workflow is structured as follows:

1. The user submits a natural language query.
2. Each LLM independently generates a structured JSON query.
3. The system performs normalization to correct issues such as invalid operators or schema mismatches.
4. The validated queries are executed on the dataset using Pandas.
5. Outputs from both models are compared.
6. A conflict resolution mechanism selects the most reliable result when discrepancies arise.

3 Key Features

The system incorporates several mechanisms to improve robustness and reliability:

- **Dual-model query generation** to reduce dependency on a single model

- **Query normalization** to correct operator and schema inconsistencies
- **Validation layer** to prevent execution of invalid queries
- **Conflict detection and resolution** between model outputs
- **Efficient data handling** through result limiting for large datasets

4 Experimental Evaluation

The system was evaluated using representative queries to analyze its performance and robustness.

4.1 Query 1: Count Rooms

Both models successfully generated structured queries for counting rooms. Minor inconsistencies, such as incorrect operators, were automatically corrected during normalization. The final outputs were consistent and accurate.

4.2 Query 2: Count Areas on Floor 716

This query highlighted differences between model outputs. One model incorrectly used an inequality operator, while the other used the correct equality condition. The system detected this discrepancy and selected the correct query based on validation rules.

These experiments demonstrate the importance of combining multiple models with structured validation mechanisms.

5 Challenges

Several challenges were encountered during development:

- Inconsistent structured outputs across different LLMs
- Incorrect operator selection (e.g., using $>$ instead of $==$)
- Misinterpretation of dataset schema
- Efficient handling of large-scale data

These challenges were addressed through normalization, validation, and conflict resolution strategies integrated into the system pipeline.

6 Conclusion

This project demonstrates that integrating multiple LLMs with structured validation significantly improves the reliability of chatbot-based data systems. Rather than relying on a single model, comparing outputs and resolving discrepancies leads to more accurate and consistent results.

A key insight is that while LLMs are effective at interpreting user intent, they require additional control mechanisms to ensure precision in structured query generation. Future work may explore more advanced validation techniques and adaptive learning strategies to further enhance system performance.